

User Guide

Beyond Mean-Variance Portfolio Optimiser

With Derivatives & Structured Products — A Mental Accounts Framework

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sami-jeddou-behavioral-portfolio-optimizer.streamlit.app

Home — choose a tool



Exact Portfolio Optimiser

Exact grid engine on the Das-Statman states
— VaR, thesis-ES and rigorous-ES, with
derivatives.



Scalable Portfolio Optimiser

Monte-Carlo scenarios + α -CVaR linear
program — scales to large, multi-derivative
portfolios.



Backtest

Out-of-sample walk-forward — expected vs
realised, plus realised alpha & beta.

Reference & navigation



About:

Methods, the mental-accounting framework and the research behind the app.



Glossary:

Plain-language, AI-powered definitions — VaR, ES, α -CVaR, copulas, alpha & beta.



Tip:

Each tile opens a tool; the Exact Portfolio Optimiser is the place to start.

Exact Portfolio Optimiser — overview



Interactive Frontiers

MV vs Behavioural vs Derivative frontier — visualised side by side on a single chart



Three Portfolios

No derivative / With derivative / Same risk — three perspectives for each optimisation run



AI-Powered

On-demand explanations for every parameter, constraint, derivative type, and result

Portfolio perspectives (0-3)

- | | | |
|---|--|--|
| 0 | Portfolio (0) — Markowitz MV optimum: | Classical mean-variance solution; Portfolio (1) coincides with it. |
| 1 | Portfolio (1) — No derivative: | Equivalent to MV optimum — confirms MVT/MAT equivalence. Most classical optimisers stop here. |
| 2 | Portfolio (2) — With derivative: | Same implied λ and downside constraint (H, α or H, α, L) — shows return uplift from derivatives. |
| 3 | Portfolio (3) — Same risk as (1): | Interpolated at $P(1)$ std dev — direct comparison of return gain at equivalent risk (indicative). |

Step 1 — Portfolio Data

1

 **PORTFOLIO DATA**

☐ Default (3-asset sample case)

☒ Live market data (Yahoo Finance)

☐ Enter manually

☐ Upload CSV

Ticker symbols (comma-separated)

AAPL, MSFT, JPM

From

To

2020/01/01

2026/06/03

Return frequency

☒ Daily ☐ Monthly

 Fetch data

Default

Das & Statman (2009) base case — 3 securities, pre-calibrated. Reproduces thesis results.

Live market data

Any global ticker from Yahoo Finance — equities, ETFs, indices, crypto (BTC, ETH...)

Enter manually

Enter means, std devs, and correlation matrix directly. Supports 2–10 securities.

Upload CSV

First col = dates, remaining cols = asset prices. Returns computed automatically.

Step 2 — Derivative / Structured Product

2 DERIVATIVE / STRUCTURED PRODUCT

Safety collar (long put + short call) ▼

None — primary securities only

Put option

Call option

Safety collar (long put + short call)

Aggressive collar (long call + short put)

Straddle (long call + long put)

Strangle (long call + long put, diff stri...)

Capital-guaranteed note — uncapped

2 DERIVATIVE / STRUCTURED PRODUCT

Safety collar (long put + short call) ▼

AI-powered: What is this instrument?

Underlying security

Sec 3 — High risk ▼

Volatility (annualised %)

50.0 - +

Default: 50.0% (std dev of Sec 3 — High risk).
Override for implied vol.

Risk-free rate (%)

3.0 - +

Maturity (years)

1.00 0.25 3.00

Put strike

1.20 0.50 1.50

Call strike

1.60 1.00 2.00

Available instruments

- ▶ Vanilla options — put, call
- ▶ Collars — safety, aggressive
- ▶ Volatility — straddle, strangle
- ▶ Spreads — bull call, bear put
- ▶ Butterfly & condor — long butterfly, call condor
- ▶ Capital-guaranteed note — capped or uncapped
- ▶ Barrier-M note & reverse convertible
- ▶ Discount & outperformance certificates



Select 'None' for no derivatives, or 'Custom' to build your own (calls, puts, digitals, ZCBs).

Step 3 — Mental-Account Constraint

3

MENTAL-ACCOUNT CONSTRAINT

AI-powered: What is the VaR constraint?

Constraint type

☒ VaR — Value at Risk
 ☐ ES — Expected Shortfall
 ☐ ES — Rigorous (beyond thesis)

Threshold H (%)

-40

-10

-1

Range extended to -40% to accommodate highly volatile assets (e.g. cryptocurrencies, emerging market equities).

Max shortfall probability α (%)

1

5

15

VaR constraint: $P(\text{return} < H) \leq \alpha$

Implied risk-aversion $\lambda = 3.6958$

MV optimal at $\lambda=3.70 \equiv$ behavioural optimal at $H=-10\%, \alpha=5\%$

VaR — Value at Risk

$$P(\text{return} < H) \leq \alpha$$

Caps the probability of finishing below H at α .

ES — Expected Shortfall

$$E[\text{return} \mid \text{return} < H] \geq L$$

Floors the average tail loss below H at L (thesis).

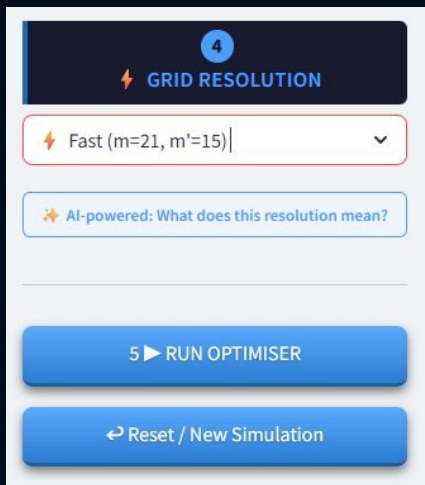
ES — Rigorous (beyond thesis)

$$E[r \mid r < H] \geq L \text{ (enforced in the solve)}$$

Enforces the ES floor in the optimisation itself.

💡 Implied λ is computed dynamically — adjust H and α to see the equivalent risk-aversion coefficient update in real time, confirming MVT/MAT equivalence.

Step 4 — Grid Resolution



The screenshot shows the 'GRID RESOLUTION' step of the optimizer. At the top, there is a blue header with a circle containing the number '4' and the text 'GRID RESOLUTION'. Below this is a dropdown menu showing 'Fast (m=21, m'=15)' with a downward arrow. Under the dropdown is a button that says 'AI-powered: What does this resolution mean?'. At the bottom of the interface are two large blue buttons: '5 ► RUN OPTIMISER' and '↶ Reset / New Simulation'.

Also available

 **Turbo — 4th resolution:** High-precision VaR accuracy in ~seconds (VaR-only, ≤ 4 assets, no derivative).

Rigorous ES: a fixed high-precision grid ($m=51$); chosen in the Risk-measure step.

Fast ($m=21, m'=15$)

 ~1–2 min (or more depending on the portfolio size)

Quick preview and parameter exploration

Standard ($m=35, m'=50$)

 ~5–15 min (or more depending on the portfolio size)

Good balance — close to thesis results

High precision ($m=51, m'=99$)

 1–2 hrs (or more depending on the portfolio size)

Publication-quality replication of thesis results

Step 5 — Run the Optimiser



Computation steps

1. Mean-variance frontier — Markowitz sweep (instant)
2. Behavioural frontier (no derivative) — State space, Gaussian copula, grid search, COBYLA
3. Derivative frontier — Same steps extended with derivative payoff (only if derivative selected)
4. Chart rendering + Results computation

Reading the Chart

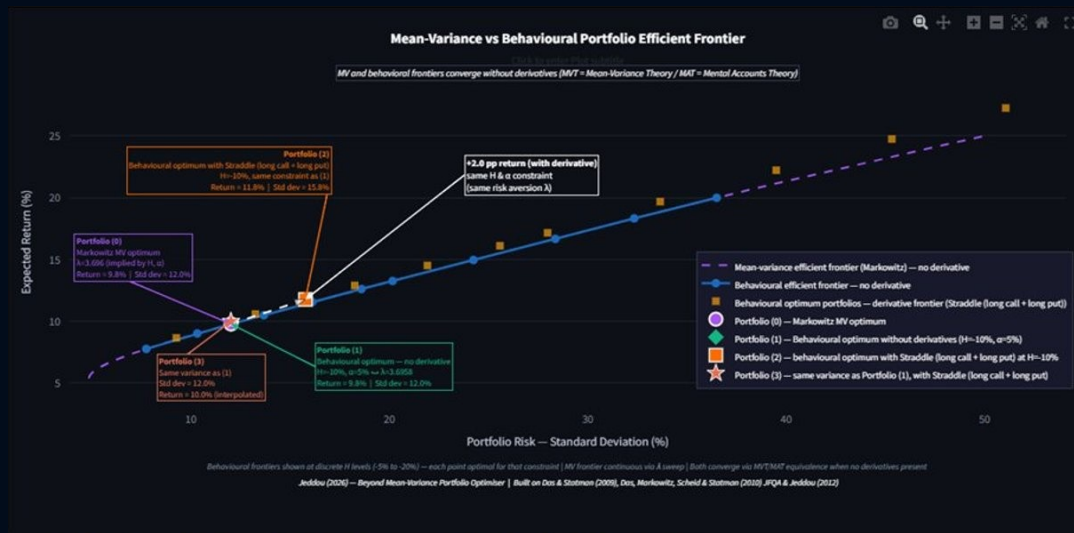
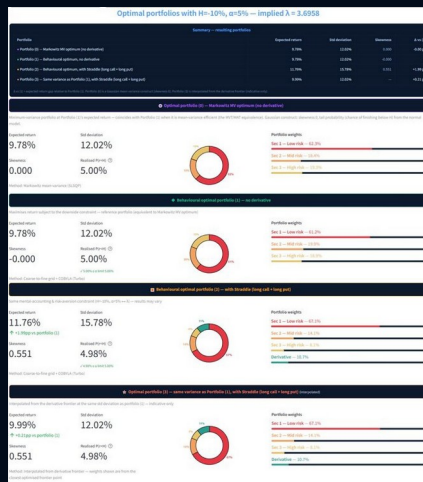


Chart legend

- Purple dashed**
Mean-variance efficient frontier (Markowitz)
- Blue dots**
Behavioural efficient frontier — no derivative
- Gold squares**
Behavioural optimum — derivative frontier
- Green diamond**
Portfolio (1) — no derivative optimum
- Orange square**
Portfolio (2) — with derivative, same λ
- Coral star**
Portfolio (3) — same variance as P(1)
- Magenta circle**
Portfolio (0) — Markowitz MV optimum

X-axis: Portfolio Risk (Standard Deviation %) | Y-axis: Expected Return (%)

Portfolio Results



Key metrics

Expected return: Annualised mean return of the optimal portfolio

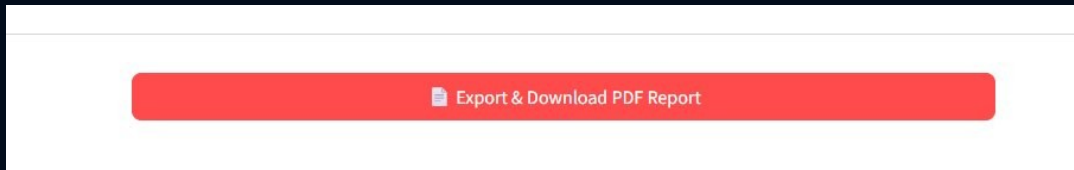
Shortfall / ES: Expected loss in the tail (ES constraint value)

Std deviation: Annualised portfolio volatility (risk measure)

Portfolio weights: Allocation to each security and derivative

Skewness: Asymmetry of return distribution — positive = upside skew

PDF Export



PDF report includes

- ✓ Cover page with run parameters
- ✓ Full efficient frontier chart
- ✓ Portfolio (1), (2), (3) metrics
- ✓ Weight allocation bar charts
- ✓ Legal disclaimer

💡 The PDF is generated in-browser — no data is sent externally. Click 'Download PDF report' after the simulation completes.

Scalable Portfolio Optimiser — Monte-Carlo + CVaR



Scenario engine

Samples thousands of joint return scenarios through a Gaussian or Student-t copula.



α -CVaR linear program

Maximises return subject to a CVaR (Expected-Shortfall) floor — Rockafellar–Uryasev.



Institutional scale

Cost grows linearly in the number of assets — dozens of holdings, several derivatives at once.

How to use it

1

Pick assets & scenarios:

Enter tickers and derivatives, choose the copula and the number of Monte-Carlo scenarios.

2

Set the ES floor:

Choose the loss level L ; the engine sweeps the return / tail-risk frontier.

3

Read the optimum:

★ marks the chosen portfolio; weights, $E[r]$, realised ES, vol and skew are reported.

Out-of-sample back-test



Two windows

Build weights on a construction window, then buy-and-hold them through a later evaluation window.



Marked to market

The derivative is repriced each step with Black-Scholes, giving a full realised return path.



Alpha & beta

Realised alpha, beta and R^2 of each holding and of the portfolio vs a chosen benchmark.

What you get

1

Expected vs realised:

Return, volatility and the loss-threshold outcome, for P1 (no derivative) and P2 (with).

2

Alpha / beta table:

Per-security and portfolio beta (the regression slope) and annualised Jensen alpha, with R^2 .

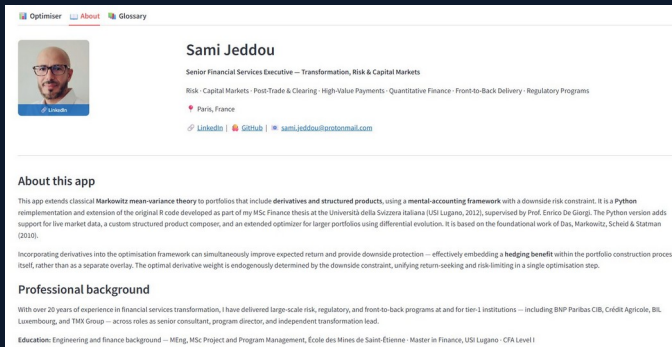
3

Optional CAPM:

Enter an expected market return $E[R_m]$ to add a CAPM required return and an ex-ante alpha.

About & Glossary Tabs

About Tab



Optimiser **About** **Glossary**

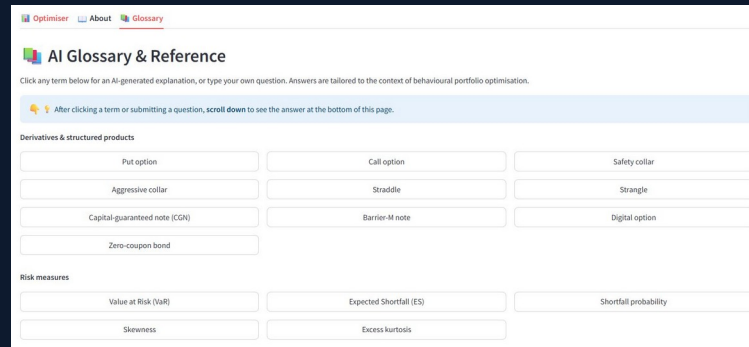
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About this app
This app extends classical Markowitz mean-variance theory to portfolios that include derivatives and structured products, using a mental-accounting framework with a downside risk constraint. It is a Python reimplementation and extension of the original R code developed as part of my MSc Finance thesis at the Università della Svizzera italiana (USI Lugano, 2012), supervised by Prof. Enrico De Giorgi. The Python version adds support for live market data, a custom structured product composer, and an extended optimizer for larger portfolios using differential evolution. It is based on the foundational work of Das, Markowitz, Scheid & Statman (2010).
Incorporating derivatives into the optimisation framework can simultaneously improve expected return and provide downside protection — effectively embedding a hedging benefit within the portfolio construction process itself, rather than as a separate overlay. The optimal derivative weight is endogenously determined by the downside constraint, unifying return-seeking and risk limiting in a single optimisation step.

Professional background
With over 20 years of experience in financial services transformation, I have delivered large-scale risk, regulatory, and front-to-back programs at and for tier-1 institutions — including BNP Paribas CIB, Crédit Agricole, BIL, Luxembourg, and THS Group — across roles as senior consultant, program director, and independent transformation lead.
Education: Engineering and finance background — MEng_MSc Project and Program Management, École des Mines de Saint-Étienne - Master in Finance, USI Lugano - CFA Level I

- ▶ Author background & contact
- ▶ App description & algorithm
- ▶ Academic references
- ▶ Legal disclaimer

Glossary Tab



Optimiser **About** **Glossary**

AI Glossary & Reference
Click any term below for an AI-generated explanation, or type your own question. Answers are tailored to the context of behavioural portfolio optimisation.
👉 After clicking a term or submitting a question, scroll down to see the answer at the bottom of this page.

Derivatives & structured products

Put option	Call option	Safety collar
Aggressive collar	Straddle	Strangle
Capital-guaranteed note (CGN)	Barrier-M note	Digital option
Zero-coupon bond		

Risk measures

Value at Risk (VaR)	Expected Shortfall (ES)	Shortfall probability
Skewness	Excess kurtosis	

- ▶ Click any term for AI explanation
- ▶ Covers derivatives, risk, portfolio theory
- ▶ Ask your own questions
- ▶ Tailored to behavioural portfolio context

Disclaimer & Contact

Legal Disclaimer

This application is based on the mental accounts portfolio optimisation framework of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010), as extended in Jeddou (2012). It is provided for educational and research purposes only and does not constitute financial advice, investment recommendations, or a solicitation to buy or sell any financial instrument. Results are purely illustrative and should not be used as the basis for any investment decision.

Academic References

- ▶ Das, S. & Statman, M. (2009) — Beyond Mean-Variance: Portfolios with Derivatives and Non-Normal Returns in Mental Accounts
- ▶ Das, S., Markowitz, H., Scheid, J. & Statman, M. (2010) — Portfolio Optimization with Mental Accounts, JFQA Vol.45 No.2
- ▶ Jeddou, S. (2012) — Beyond Mean-Variance: Options and Structured Products in Behavioral Portfolios, MSc Finance, USI Lugano